Credit Card Fraud Risk Analysis

Power BI + DAX + Python | 1,000 transactions | Dec 2023 - Dec 2024

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28.6%
fraud rate: 286 of 1,000 transactions

INR 33.1 L
fraud value, 27.0% of INR 1.22 crore transacted

50.9%
of fraud value comes from Card Not Present and Identity Theft

9.8%
of confirmed fraud was rated Critical; 39.5% was rated Low

Executive Summary

Banks lose money and customer trust when fraudulent card transactions slip through. I analyzed **1,000 credit card transactions (Dec 2023 - Dec 2024)** in Python and built an interactive **Power BI dashboard with six custom DAX measures** to show how much fraud there is, what types dominate, where and when it happens, and whether the existing risk labels actually catch it.

> **Bottom line:** fraud is widespread (**28.6%** of transactions, **INR 33.1 lakh**) and spread fairly evenly across states, cards, banks and merchants, but the **risk labels and fraud scores do not separate fraud from genuine activity**: Critical-rated transactions are fraudulent 26.2% of the time, less often than Low-rated ones (28.2%). Recalibrating the scoring rules is the biggest opportunity, with Card Not Present and Identity Theft controls as the first priority.

Key Findings at a Glance

- **Fraud types:** Card Not Present (72 transactions) and Identity Theft (64) are the largest and together hold 50.9% of fraud value (INR 16.8 lakh).
- **Risk labels:** 39.5% of confirmed fraud sits in the Low bucket and only 9.8% in Critical; fraud score averages 53.9 for fraud vs 52.9 for genuine (no meaningful gap).
- **Where:** Maharashtra, Karnataka and Rajasthan have the most fraud (36.4% together); by merchant city, Ahmedabad (36.7%), Mumbai and Chennai run far above Delhi (18.9%), the only statistically significant gap found ($p = 0.04$).
- **When:** the fraud rate rose from 25.7% in the first half of the year to 31.7% in the second half ($p = 0.04$), peaking in June and August-September 2024.
- **Statistical honesty:** across states, cards, banks, merchants and categories, most rate gaps are within sampling noise (all $p > 0.1$), so they are signals to monitor, not proven drivers.

Business Problem

The fraud team needs to know **how much fraud is happening, which fraud types and segments matter most, and whether the current risk classification is doing its job.** The dashboard gives investigators a single view; this report adds the numbers behind it.

Tools & Techniques

Area	What I used
Power BI + DAX	One-page dashboard with 6 custom DAX measures, slicers and linked visuals (details in section 1)
Python	Pandas, SciPy: profiling, group-level analysis, confidence intervals and chi-square tests

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1. Data & Power BI Model

Item	Detail
Dataset	1,000 transactions x 14 columns: ID, customer, merchant, date, amount (INR), fraud risk label, fraud type, state, card type, bank, IsFraud flag, fraud score, category, merchant location
Quality checks	No missing values; 1,000 unique transaction IDs; no duplicates; amounts INR 141 to INR 24,993; dates 16 Dec 2023 to 15 Dec 2024 (first and last months are partial)
Power Query	CSV import, header promotion, and data types set for all 14 columns (dates as Date, amounts and flags as whole numbers)
Note	Fraud Type is filled even for genuine transactions, so every fraud-type analysis here uses only IsFraud = 1 rows, exactly as the DAX measures do

DAX measures used in the dashboard

The model is a single table, `Credit Card Fraud Risk Analysis`, with six custom measures. I verified each result against the raw CSV.

Measure	DAX (abbreviated)	What it does	Result
Total Transaction Amount	`SUM(Table[Transaction Amount (INR)])`	Total value of all transactions in the current filter	INR 12,226,843
Fradulent Transaction	`CALCULATE(COUNTROWS(Table), Table[IsFraud] = 1)`	Count of fraud-flagged transactions	286
Fraud Rate %	`DIVIDE([Fradulent Transaction], COUNTROWS(Table), 0) * 100`	Share of transactions that are fraud	28.6
Critical Risk Transaction	`DIVIDE(CALCULATE(COUNTROWS(Table), Table[Fraud Risk] = "Critical"), COUNTROWS(Table), 0) * 100`	Share of transactions rated Critical (a percentage, not a count)	10.7
Fraudulent Transaction Amount	`CALCULATE([Total Transaction Amount], Table[IsFraud] = 1)`	Total value lost to fraud-flagged transactions	INR 3,306,813
Top Fraud Type	`VAR Toptype = TOPN(1, SUMMARIZE(Table, Table[Fraud Type], "Fraudulent Count", CALCULATE(COUNTROWS(Table), IsFraud = 1)), [Fraudulent Count], DESC) RETURN MAXX(Toptype, Table[Fraud Type])`	Fraud type with the most fraud-flagged transactions (dynamic: changes with slicers)	Card Not Present

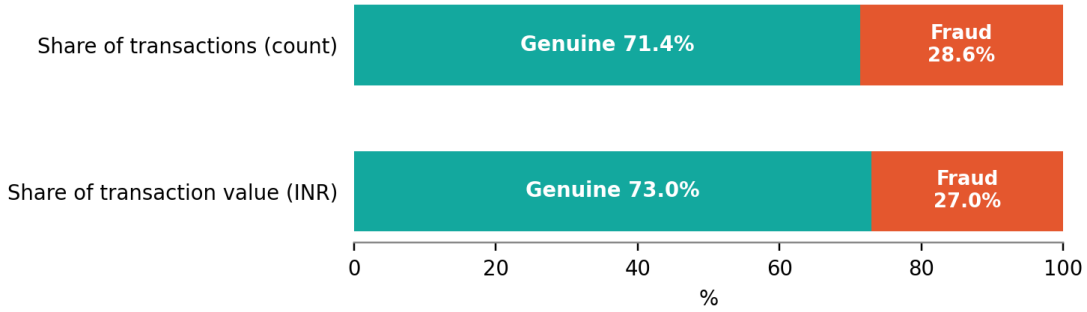
Model review

- **Measures chain together:** Fraud Rate % reuses Fradulent Transaction, and Fraudulent Transaction Amount reuses Total Transaction Amount, so one definition feeds every visual.
- **Naming fixes:** "Fradulent" is a typo (Fraudulent), and "Critical Risk Transaction" returns a percentage, so "Critical Risk %" is clearer. Renaming updates visuals automatically.
- **Dates:** there is no custom date table; Power BI's automatic date hierarchy (Year, Quarter, Month, Day) is used by the monthly trend line.

2. Fraud Overview & Fraud Types

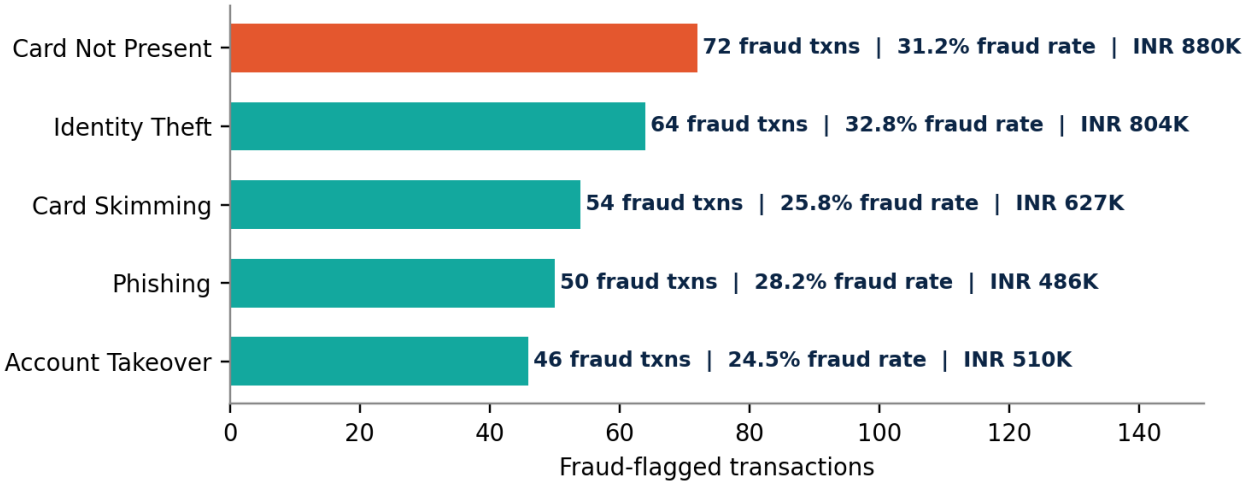
Finding 1: Nearly three in ten transactions are fraud

286 of 1,000 transactions are flagged as fraud (95% confidence interval 25.9% to 31.5%), worth **INR 33.07 lakh** of INR 1.22 crore. The average fraud transaction (INR 11,562) is slightly smaller than a genuine one (INR 12,493), though that gap is not statistically significant (p = 0.06).



Context: a 28.6% fraud share is far higher than real-world card portfolios, so this looks like a curated or sampled dataset. The figures describe this sample, and are best used to demonstrate the analysis and dashboard rather than to size real-world fraud.

Finding 2: Card Not Present and Identity Theft lead

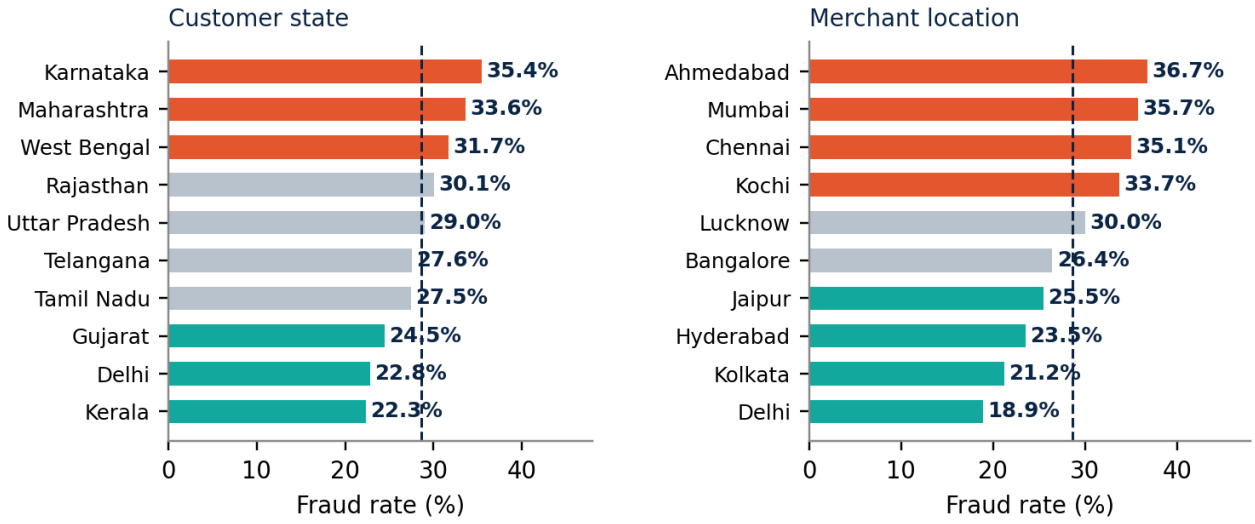


Fraud type	Fraud txns	Share of fraud txns	Share of fraud value	Avg fraud amount (INR)
Card Not Present	72	25.2%	26.6%	12,218
Identity Theft	64	22.4%	24.3%	12,568
Card Skimming	54	18.9%	19.0%	11,620
Account Takeover	46	16.1%	15.4%	11,079
Phishing	50	17.5%	14.7%	9,713

Card Not Present and Identity Theft hold **50.9% of fraud value**. Phishing cases are the smallest on average (INR 9,713). Differences in fraud rate between types (24.5% to 32.8%) are not statistically significant (p = 0.31), so the volume and value ranking matters more than the rate ranking.

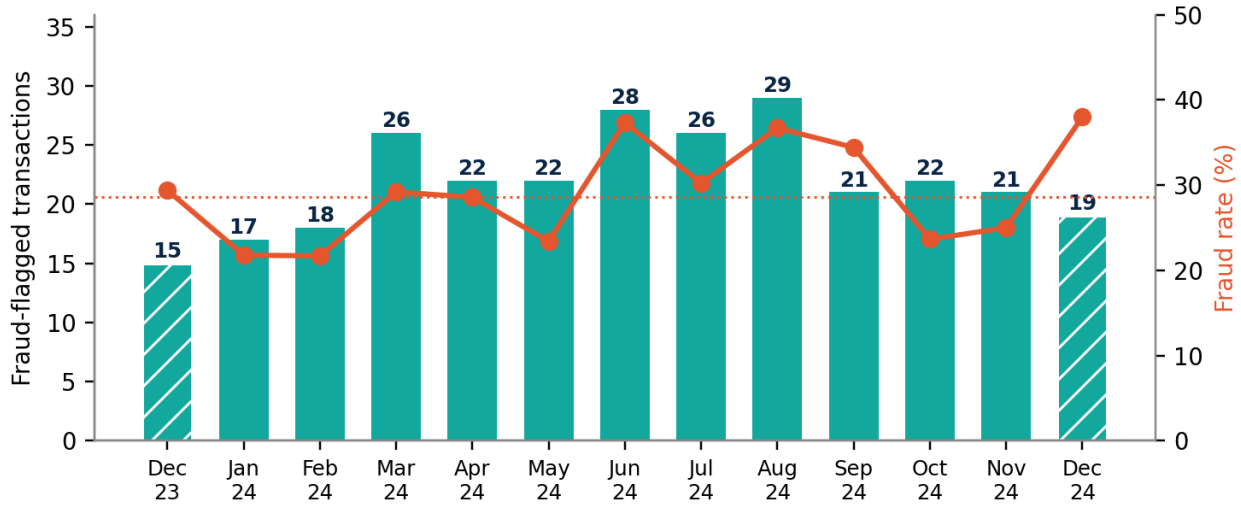
3. Where and When Fraud Happens

Finding 3: Geography



- **By customer state**, Maharashtra (36), Karnataka (34) and Rajasthan (34) have the most fraud transactions, 36.4% of the total. Karnataka (35.4%) and Maharashtra (33.6%) also have the highest rates; Kerala (22.3%) and Delhi (22.8%) the lowest. These gaps are not statistically significant (p = 0.50).
- **By merchant location**, Ahmedabad (36.7%), Mumbai (35.7%), Chennai (35.1%) and Kochi (33.7%) run well above the 28.6% average, while Delhi (18.9%) and Kolkata (21.2%) run well below. This is the **only dimension with a statistically significant gap (p = 0.039)**, though with nine comparisons run, one borderline result can occur by chance, so it should be confirmed on more data.

Finding 4: Fraud rose in the second half of the year



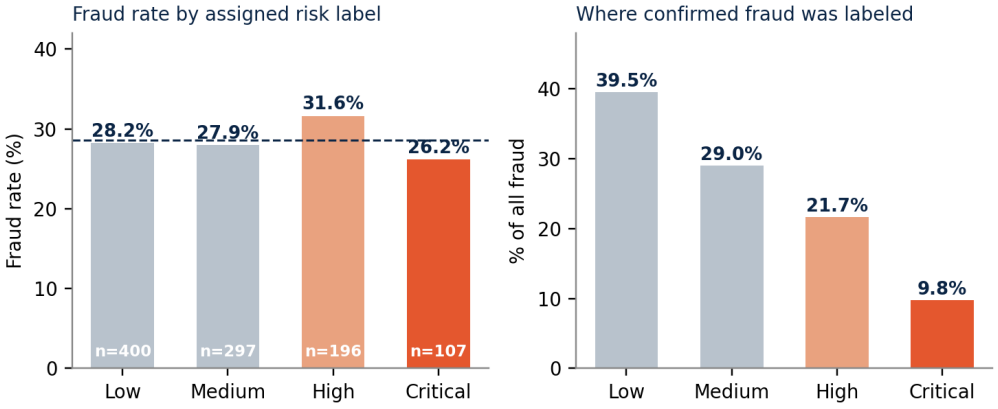
Hatched bars are partial months (16-31 Dec 2023 and 1-15 Dec 2024). The dotted line is the 28.6% average.

- Fraud-flagged transactions peak in **June (28), August (29)** and March (26); January and February are the quietest (17-18).
- The fraud **rate rises from 25.7% (Dec 2023 to mid-Jun 2024) to 31.7% (mid-Jun to Dec 2024)**, a statistically significant increase (p = 0.042), with monthly rates up to 37-38% in June, August and December.

> **Business insight:** a sustained second-half rise justifies a monthly alert on the Fraud Rate % measure and a review of controls introduced or relaxed around mid-year.

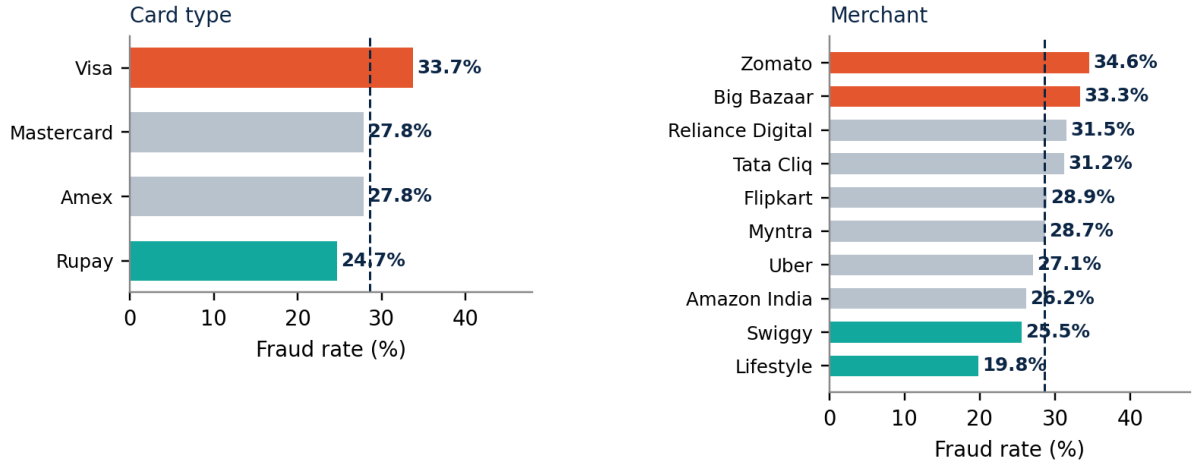
4. Do the Risk Labels Work? Segment Comparison

Finding 5: The assigned risk labels do not separate fraud from genuine transactions



Business insight: if the labels worked, fraud rates would climb from Low to Critical. Instead they are **28.2%, 27.9%, 31.6% and 26.2%** (chi-square p = 0.73), and Critical-rated transactions (10.7% of volume, INR 13.3 lakh) are fraudulent less often than Low-rated ones. Fraud score shows the same pattern (53.9 for fraud vs 52.9 for genuine, p = 0.55). **Only 28 of 286 confirmed frauds were rated Critical; 113 were rated Low.**

Finding 6: Card, merchant and other segments



Visa (33.7%) and Zomato (34.6%) lead; Rupay (24.7%) and Lifestyle (19.8%) trail. Which differences are strong enough to trust?

Dimension	Highest fraud rate	Lowest fraud rate	Chi-square p	Verdict
Risk label	High 31.6%	Critical 26.2%	0.73	No signal
Fraud type	Identity Theft 32.8%	Account Takeover 24.5%	0.31	Not significant
Customer state	Karnataka 35.4%	Kerala 22.3%	0.50	Not significant
Card type	Visa 33.7%	Rupay 24.7%	0.15	Not significant
Bank	Andhra Bank 34.7%	Yes Bank 25.4%	0.86	Not significant
Category	Food Delivery 32.4%	Transportation 24.6%	0.60	Not significant
Merchant	Zomato 34.6%	Lifestyle 19.8%	0.43	Not significant
Merchant location	**Ahmedabad 36.7%**	**Delhi 18.9%**	**0.039**	**Significant (borderline)**

5. Dashboard, Recommendations & Next Steps

Power BI dashboard

A one-page interactive report (13 elements) lets investigators filter everything by fraud type, state and merchant.

Component	Content
Multi-row card	Fraud Rate %, Fradulent Transaction, Critical Risk Transaction, Fraudulent Transaction Amount and Top Fraud Type (all six measures drive the page)
Bar chart	Total transaction amount by fraud type and transaction category
Donut chart	Total transaction amount by fraud risk label
Column chart	Fraud-flagged transactions by state
Line chart	Fraud-flagged transactions by month
3 slicers	Fraud type, state and merchant name

Recommendations

#	Action	Evidence
1	**Recalibrate the risk-scoring rules** so labels track confirmed fraud; test new rules against the IsFraud outcome	Critical 26.2% vs Low 28.2% fraud rate; only 9.8% of fraud rated Critical
2	**Strengthen Card Not Present and identity-verification controls** (step-up authentication, device checks)	50.9% of fraud value; 136 of 286 fraud transactions
3	**Review merchants in Ahmedabad, Mumbai, Chennai and Kochi**, and validate on more data	Rates 33.7-36.7% vs Delhi 18.9% (p = 0.039)
4	**Set a monthly alert on Fraud Rate %** and investigate the second-half rise	25.7% to 31.7% (p = 0.042)

Limitations

- A 1,000-row sample with an unusually high fraud share; Fraud Type is populated for genuine transactions, so only IsFraud = 1 rows are used for type analysis.
- Most segment differences are within sampling noise, and the first and last months are partial. The dashboard's Month-level axis would merge the two Decembers (34 fraud transactions), so use a Year-Month axis.

Next steps

- Add measures: `Fraud Value % = DIVIDE([Fraudulent Transaction Amount], [Total Transaction Amount], 0)`, an Average Fraud Amount, and a "fraud rated Low" share to monitor missed fraud.
- Add a fraud-rate-by-risk-label visual and a Year-Month trend to the dashboard.

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Code and dashboard: https://github.com/jeeruvicky51-bit/Credit-Card-Fraud-Risk-Analysis